Scientific Research Methodology
4. Theory of knowledge, research principles

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Epistemology: the theory of knowledge

- ⋆ a branch of philosophy
- ⋆ the study of the nature, origins, and limits of knowledge
- ⋆ a fundamental tool for acquiring knowledge: **the question**
  - questions about oneself
  - questions addressed to others
  - questions directed at objects and aspects of the physical world
- ⋆ the purpose of inquiry in scientific research: **finding the truth**

What is truth?

⋆ from a scientific perspective
⋆ **mathematics:** a result that can be obtained through fully rigorous logical reasoning, starting from a set of axioms or postulates
⋆ **applied mathematics**: using mathematical concepts and simulations to address a practical need/question
⋆ **physics**: a theory that explains how phenomena unfold in nature

The historical perspective

- **faith**: people attributed anything beyond explanation to a deity
- **science**: people sought to understand the essence of things, living beings, and natural phenomena
  - A fundamental element: putting received ideas to the test
  - No theory is final
  - A scientific result has practical value

Knowledge vs. learned ignorance

- organized vs. disorganized
- accepts supervision and a role model vs. self-taught, with no contacts in the field
- academic affiliation vs. independence
- obtaining **true** results vs. obtaining results at any cost
- truth, order, accuracy vs. falsehood, sensationalism, pseudoscience
- belonging to a school of thought; scientific identity vs. independence, with no scientific identity

[Ristea, Ioan-Franc, Popescu, *Methods in Scientific Research*]

Everyday knowledge vs. scientific knowledge

⋆ superficiality vs. depth

Scientific knowledge

- empirical knowledge: developing sound intuitions, observation, representations
- theoretical knowledge: the result of thought processes

Table 1.2. Everyday knowledge versus scientific knowledge

Distinguishing features	Everyday knowledge	Scientific knowledge
1. In terms of **form**	It has no theoretical basis, forming a heterogeneous body of practical knowledge acquired through learning by doing, under the pressure of everyday needs.	It takes shape and is disseminated in the form of scientific theories.
2. In terms of **method**	It does not arrive at rigorous, verifiable truths.	It has a methodological perspective, organized and developed through the use of rigorous methods.
3. In terms of **hypothesis testing procedures**	It has only one way of testing hypotheses: trial and error.	It uses complex criteria, techniques, and tests to verify truth.
4. In terms of **language**	It uses ordinary language.	It uses specialized, highly abstract language that differs from one scientific discipline to another.

Source: Table based on Gh. Răboacă and D. Ciucur (2001, pp. 50–51).

1. The principle of competence

[C. Enăchescu, *Treatise on the Theory of Scientific Research*, 2007]
[Ristea, Ioan-Franc, Popescu, *Methods in Scientific Research*, 2017]

a) to be a researcher, one must be a trained specialist in the relevant scientific field
b) a researcher has special qualities: a desire for knowledge, curiosity, critical thinking, passion, patience, perseverance
c) a preliminary stage: gaining experience in the specialty

2. The principle of defining the research objective

a) the object of research: setting a precise objective
b) deviations from the objective are not permitted during the research
c) the object of research must not be modified to suit the available methods and techniques; these must be adapted to it!

Remark: A researcher with a certain degree of maturity will know how to adapt the objective if necessary

3. The principle of truth

The aim of scientific research is to discover the truth about the object on which the research focuses.

a) The aim is to find the truth: not to obtain a desired result at any cost
b) research is conducted logically and coherently
c) research results are expressed in clear and precise language
d) scientific research is based on logical reasoning (without subjectivity or speculation)

4. The principle of method

a) a rigorous plan, established a priori
b) coherence in the logical process of discovering the truth
c) working techniques and methods are adapted to the nature of the object under study
d) research proceeds from the simple to the complex

5. The principle of demonstration

a) scientific work is demonstrated and verified
b) research leads to a theoretical model that relates the theoretical model to its properties
c) valid and reliable research results are incorporated into the body of knowledge of the relevant scientific field.

6. The principle of correlation

a) connections are established between the object under study and related objects
b) the properties of related objects are taken into account in the study
c) the results obtained will be useful for similar objects in the field

7. The principle of evaluating results

a) All research results are evaluated correctly, on strictly rational grounds
b) Results are compared with those already available in the scientific literature

8. The principle of utility

a) research results should be accessible and usable by other specialists as soon as possible (in theory and in practice)
b) every scientific research result has an element of originality within its scientific field

9. Scientific and moral responsibility

a) standards of conduct, principles of professional ethics

- confidentiality
- respect for human dignity
- impartiality
- the overriding interest
- professional independence

b) collaboration with colleagues in related fields is accepted; duplication of studies and results is avoided

c) the researcher defends the results against criticisms that may be raised about the research undertaken

Examples

Example 1

Given the sides a, b, c of an acute triangle, what is the optimal constant $k > 0$ in the inequality below?

$$a^2 + b^2 + c^2 \leq k[(a-b)^2 + (b-c)^2 + (c-a)^2] + 4\sqrt{3}S.$$

Example 2

For a polynomial $f(z) = (z - r_1)(z - r_2)...(z - r_n)$, $n \geq 2$, whose roots $r_1, ..., r_n$ lie in the unit disk $|z| \leq 1$, each root is at a distance of at most 1 from a critical point.